A Multi-Threaded Broken-Link Checker for the Marian University Website

An Operating-Systems–Informed Design and Evaluation

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BACKGROUND & MOTIVATION

INTERNSHIP

Startet in May in Marketing & Communications

WEB TEAM

worked on large-scale projects

MAJOR TASK

Full website migration (~1,600 pages)

ISSUES

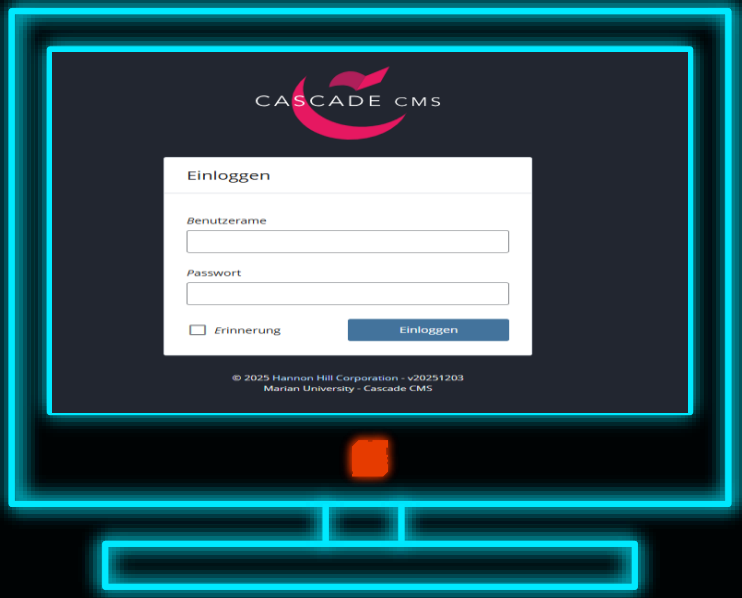
broken links, missing redirects, outdated content

IDEA

tool to automate link checking

WHY THIS PROJECT MATTERS

- Website migration revealed broken links & Cascade login links
- ~1,600 pages → impossible to check manually
- Issues often found by accident
- Departments frequently request help fixing links
- Time-consuming and inefficient process
- My crawler:
  - Automatically scans all pages
  - Detects broken + Cascade login links
  - Shows exactly where problems appear
  - Provides clear reports for fast fixes



CASCADE LOGIN LINK

- Some public pages accidentally link to internal admin login pages
- These pages should only be visible to the Web Team
- For users, seeing this link = broken / incorrect link
- My crawler detects these links the same way as broken links

LINK SUMMARY

N21

	A	B	H	K	L
1	url	status	internal_links_found	violation	violations_count
2	https://www.marian.edu/_beacon	200	1	none	0
3	https://www.marian.edu/why-mu	200	2	none	0
4	https://www.marian.edu/faith	200	3	none	0
5	https://www.marian.edu/visita	200	3	none	0
6	https://www.marian.edu/fafsa	200	7	none	0
7	https://www.marian.edu/community	200	8	none	0
8	https://www.marian.edu/alumni	200	17	none	0
9	https://www.marian.edu/families	200	7	none	0
10	https://www.marian.edu/blog	200	36	none	0
11	https://www.marian.edu/apply-now	200	6	none	0
12	https://www.marian.edu/libraries	200	6	cascade_login	2
13	https://www.marian.edu/online	200	17	none	0
14	https://www.marian.edu/plymouth	200	13	none	0
15	https://www.marian.edu/athletics	200	5	none	0
16	https://www.marian.edu/library	200	18	none	0
17	https://www.marian.edu/the-exchange	200	4	none	0
18	https://www.marian.edu/provost	200	24	none	0
19	https://www.marian.edu/academics	200	25	none	0
20	https://www.marian.edu/privacy-policy	200	1	none	0
21	https://www.marian.edu/about-marian	200	3	none	0
22	https://www.marian.edu/campus-life	200	14	none	0
23	https://www.marian.edu/search-results	200	1	none	0
24	https://www.marian.edu/earn-and-learn	200	8	cascade_login	2
25	https://www.marian.edu/404	301	33	none	0
26	https://www.marian.edu/human-resources	200	1	none	0

RUN SUMMARY

I15

	A	B	C	D	E	F	G	H	I	J	K
1	ts_utc	scheduler	threads	delay_s	urls_total	duration_s	urls_per_s	broken_links_	cascade_logins_total	pages_with_violations	total_links_found
2	2025-11-19T20:24:30Z	priority	12	0.5	1618	175,88	9,2	26	335	95	12258
3	2025-11-20T19:07:05Z	priority	12	0.5	1618	523,78	3,09	27	335	97	12245
4	2025-11-20T19:16:45Z	priority	12	0.5	1618	543,01	2,98	27	335	96	12245
5	2025-11-20T19:24:10Z	fifo	12	0.5	1618	406,57	3,98	27	335	97	12245
6	2025-11-20T19:30:20Z	fifo	12	0.5	1618	357,6	4,52	26	335	96	12245
7	2025-11-20T19:40:13Z	fifo	12	0.5	1618	569,6	2,84	32	335	102	12241
8	2025-11-20T20:00:06Z	fifo	12	0.5	1618	345,34	4,69	29	335	98	12245
9	2025-11-20T20:14:47Z	priority	12	0.5	1618	437,07	3,7	39	335	104	12245
10	2025-11-21T15:35:05Z	priority	12	0.5	1618	265,58	6,09	25	335	95	12245
11	2025-11-26T03:55:45Z	priority	12	0.5	1618	330,9	4,89	26	324	95	12240
12	2025-11-26T17:20:30Z	priority	12	0.5	1618	183,76	8,8	29	323	94	12240
13	2025-12-01T15:03:03Z	priority	12	0.5	1618	186,22	8,69	29	323	96	12241
14											

VIOLATION SUMMARY

	A	B	C	D	E	F	G
1	page_url	link_url	violation_type	status	final_url	note	
2	https://www.marian.edu/libraries	https://marian.cascadecms.com/entity/open.act	cascade_login			cascade login link	
3	https://www.marian.edu/libraries	https://marian.cascadecms.com/entity/open.act	cascade_login			cascade login link	
4	https://www.marian.edu/earn-and-learn	https://marian.cascadecms.com/entity/open.act	cascade_login			cascade login link	
5	https://www.marian.edu/earn-and-learn	https://marian.cascadecms.com/entity/open.act	cascade_login			cascade login link	
6	https://www.marian.edu/dean-of-students	https://connect.marian.edu/submitter/form/start/558272	broken_link	404	https://conne	status>=400	
7	https://www.marian.edu/consumer-informa	https://marian.cascadecms.com/entity/open.act	cascade_login			cascade login link	
8	https://www.marian.edu/consumer-informa	https://marian.cascadecms.com/entity/open.act	cascade_login			cascade login link	
9	https://www.marian.edu/consumer-informa	https://marian.cascadecms.com/entity/open.act	cascade_login			cascade login link	
10	https://www.marian.edu/consumer-informa	https://marian.cascadecms.com/entity/open.act	cascade_login			cascade login link	
11	https://www.marian.edu/consumer-informa	https://marian.cascadecms.com/entity/open.act	cascade_login			cascade login link	
12	https://www.marian.edu/consumer-informa	https://marian.cascadecms.com/entity/open.act	cascade_login			cascade login link	
13	https://www.marian.edu/consumer-informa	https://marian.cascadecms.com/entity/open.act	cascade_login			cascade login link	
14	https://www.marian.edu/consumer-informa	https://marian.cascadecms.com/entity/open.act	cascade_login			cascade login link	
15	https://www.marian.edu/consumer-informa	https://marian.cascadecms.com/entity/open.act	cascade_login			cascade login link	
16	https://www.marian.edu/consumer-informa	https://marian.cascadecms.com/entity/open.act	cascade_login			cascade login link	
17	https://www.marian.edu/consumer-informa	https://marian.cascadecms.com/entity/open.act	cascade_login			cascade login link	
18	https://www.marian.edu/consumer-informa	https://marian.cascadecms.com/entity/open.act	cascade_login			cascade login link	
19	https://www.marian.edu/consumer-informa	https://marian.cascadecms.com/entity/open.act	cascade_login			cascade login link	
20	https://www.marian.edu/consumer-informa	https://marian.cascadecms.com/renderfile/5b83941d0a(	cascade_login			cascade login link	
21	https://www.marian.edu/consumer-informa	https://marian.cascadecms.com/entity/open.act	cascade_login			cascade login link	
22	https://www.marian.edu/consumer-informa	https://marian.cascadecms.com/entity/open.act	cascade_login			cascade login link	
23	https://www.marian.edu/consumer-informa	https://marian.cascadecms.com/entity/open.act	cascade_login			cascade login link	
24	https://www.marian.edu/consumer-informa	https://marian.cascadecms.com/renderfile/5b83941d0a(	cascade_login			cascade login link	
25	https://www.marian.edu/consumer-informa	https://marian.cascadecms.com/entity/open.act	cascade_login			cascade login link	
26	https://www.marian.edu/consumer-informa	https://marian.cascadecms.com/entity/open.act	cascade_login			cascade login link	
27	https://www.marian.edu/consumer-informa	https://marian.cascadecms.com/entity/open.act	cascade_login			cascade login link	

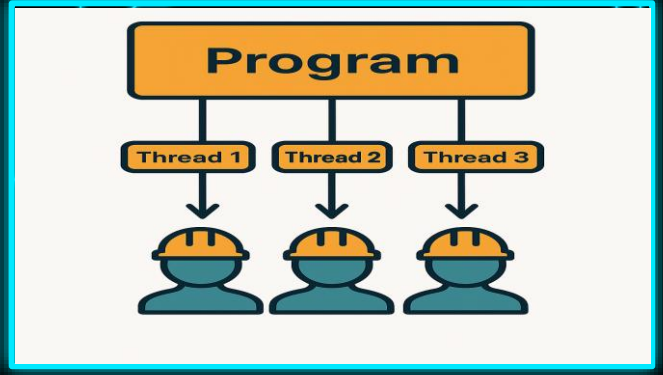
SYSTEM OVERVIEW

3 main components

- Scheduler: loads sitemap & queues URLs
- Worker Threads: process pages in parallel
- Cache Layer: avoids repeated HTTP requests

Config via YAML

- Choose FIFO or Priority mode
- Enable/disable cache
- Easy for the web team to adjust



WHY THREADS?

- Threads = multiple workers running in parallel
- Great for I/O-bound tasks like HTTP requests
- Allows:
  - Multiple pages checked at once
  - Network delay hidden
  - Faster overall crawl
- Performance:
  - 1 thread → ~40 min
  - 12 threads → 3–5 min

LRU CACHE (LEAST RECENTLY USED)

- Stores results of previously checked URLs
- Avoids re-downloading the same page
- Benefits:
  - Saves network traffic
  - Speeds up the crawl
  - Prevents redundant checks
- Removes the least recently used item when full

```python
while True:
    try:
        url = frontier.get_nowait()
    except queue.Empty:
        break

    # Duplicate-Schutz
    with vlock:
        if url in visited:
            frontier.task_done()
            continue
        visited.add(url)

    _rate_limit(delay)

    t_name = threading.current_thread().name
    ts_start = datetime.utcnow().isoformat(timespec="seconds") + "Z"
    t0 = perf_counter()
```

```python
resp = session.get(url, headers=headers, timeout=timeout, allow_redirects=True)

if extract and resp.ok and "text/html" in ctype.lower():
    link_info = _extract_internal_links(...)
    links_for_page = link_info["links"]
```

WORKER LOGIC

- Takes URL from shared queue
- Checks if page was already visited
- Sends rate-limited HTTP request
- Extracts internal links from HTML
- Checks each link (Cascade login / broken link)
- Records violations + page summary
- Thread-safe writing using locks

FUTURE WORK

PDF SCANNING

- Detect outdated content
- Identify PDFs

WEB APPLICATION

- Run crawler without my device

CSV REPORTS

- Automatic CSV reports for the whole web team

MONITORING

- Potential for full website quality monitoring tool

THANKS!

Do you have any questions?